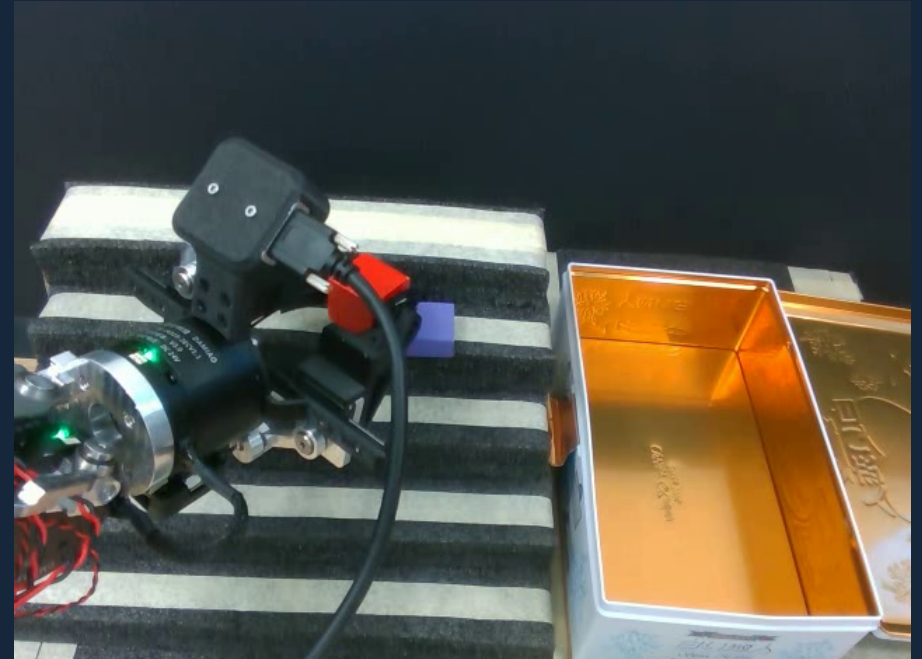


CAST

Anchoring critic-guided fine-tuning
of generative robot policies

**Improve the skill.
Keep what works.**

Persistent reference + useful adaptation



OpenArm · staircase manipulation

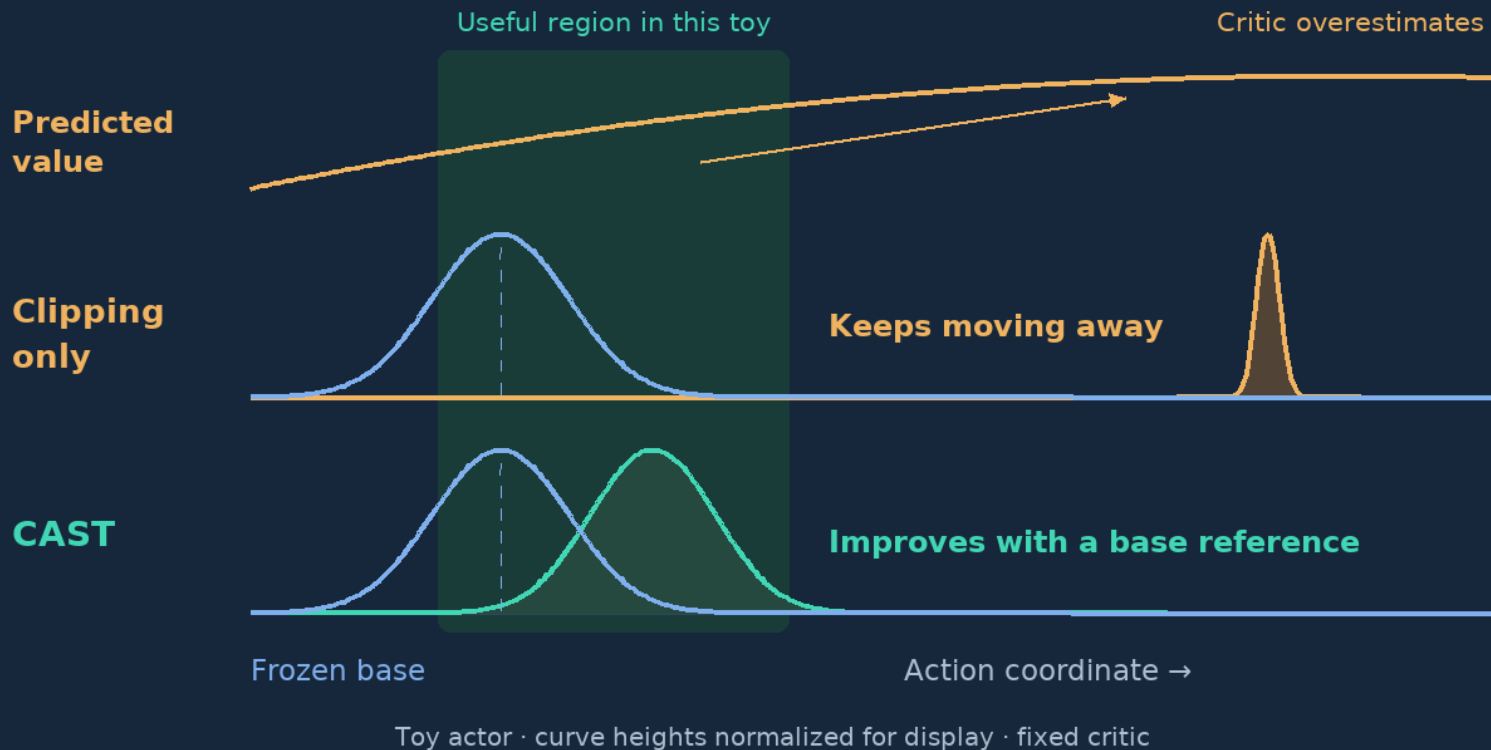
Teleoperated demonstration frame

Small steps can still drift.

An imperfect critic can turn local corrections into cumulative drift.

Small steps can accumulate into large drift.

Same starting policy. Same critic. Same step cap.



Clipping

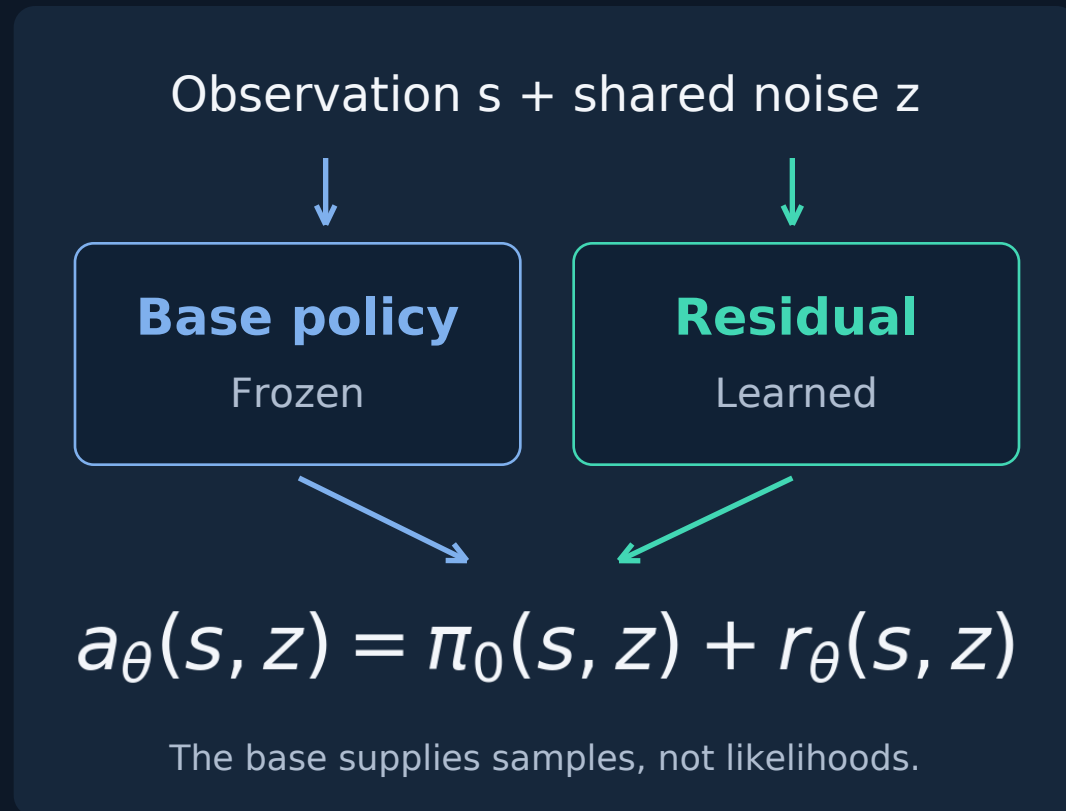
Limits each individual correction.

Restoration

Responds to departure already accumulated.

Freeze the base. Learn a correction.

Use the same observation and noise to pair the base and current action.



Every action has its own frozen reference.

The same noise sample goes through both branches.

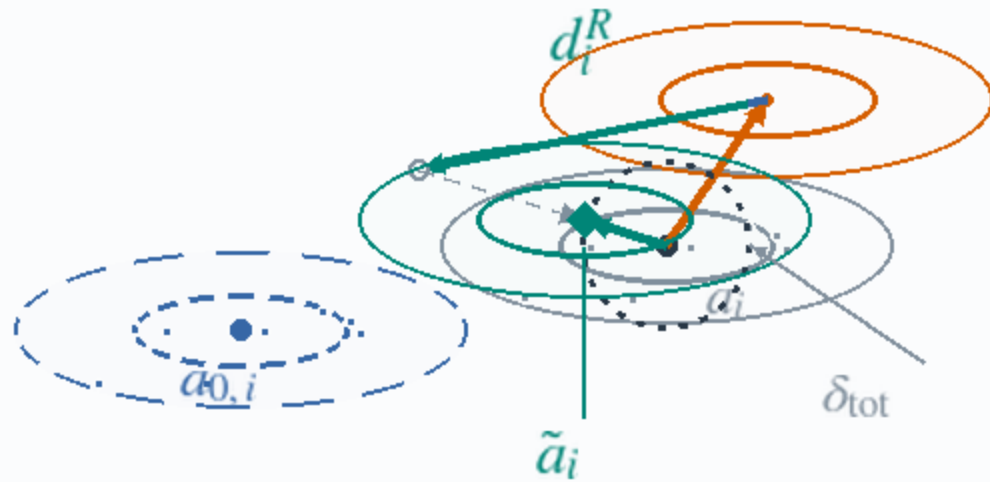


Current action = frozen base action + learned residual

One observation, three noise samples · references stay paired during training

3 Restore + clip

Cap around the current action



Limit the combined target step

$$\Delta_i = C_{\delta_{\text{tot}}}(d_i^Q + d_i^R)$$

$$\tilde{r}_i = \text{sg}[\bar{r}_i + \Delta_i]$$

$$\|\tilde{r}_i - \bar{r}_i\|_2 \leq \delta_{\text{tot}}$$

The circle limits this target step.

Its center is the current action.

Same numerical construction as paper Figure 2 · illustrative action space

CAST / ICRA 2027 supplementary video

A soft reference, learned through fixed targets.

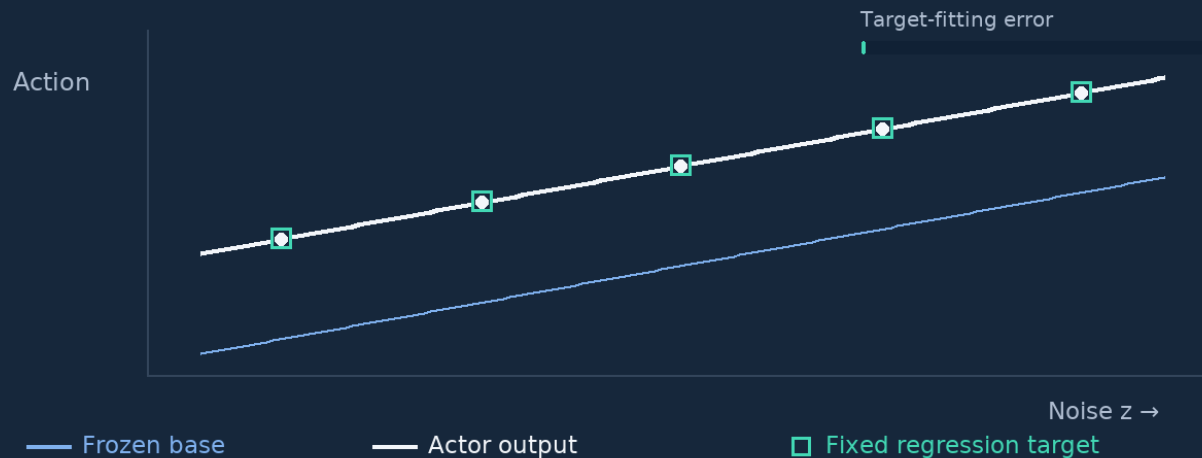
The target is fixed. The actor learns to match it.

Update the residual parameters, then construct fresh targets.

Construct targets

Fit the residual

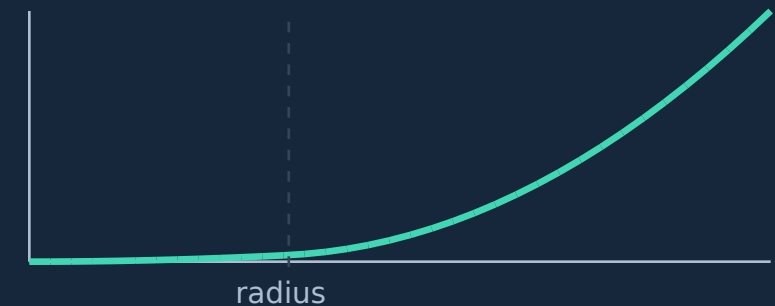
Refresh targets



Actual gradient-descent updates of a small residual actor · targets are detached

Restoration penalizes departure from the base.

Illustrative penalty $U(r)$



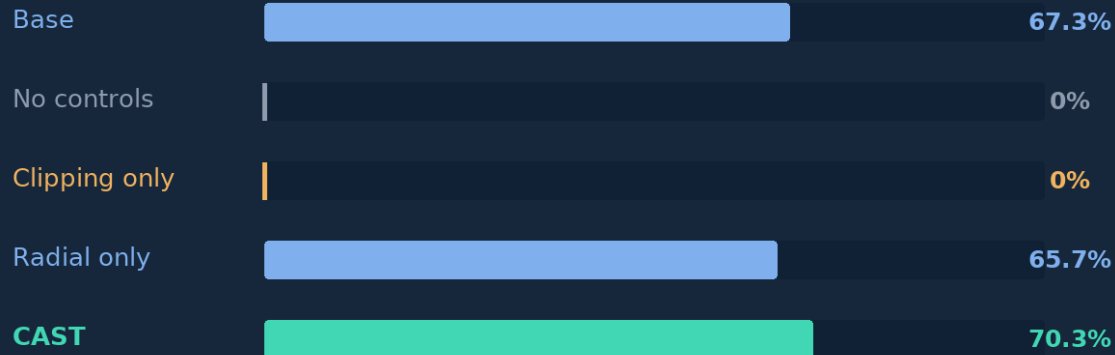
Soft threshold, not a hard wall

With inactive caps: the same local gradient direction as regularized critic optimization.

Reach useful performance with less experience.

Restoration matters on the tested tasks.

LIBERO · mean success across five tasks · measured results



Equal comparison: all five update configurations are shown.

Reported aggregate success · not a success-versus-training curve

Steps to 90% success

CAST



149.3K

DICE-RL

240.0K

38%

fewer environment
steps on average

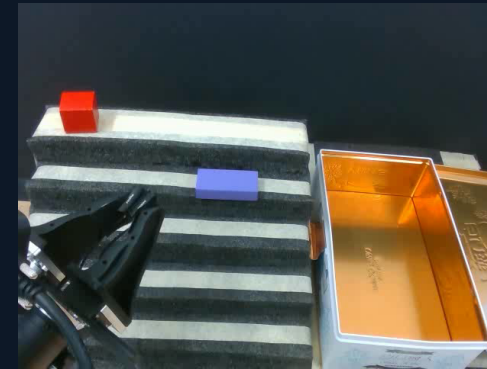
Shared diffusion base · three runs

LIBERO: clipping without restoration loses all success on five tested tasks.

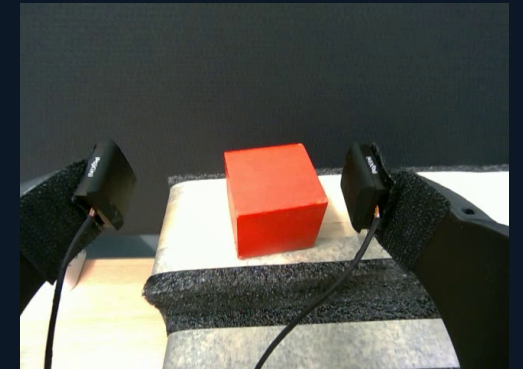
From recorded experience to robot execution.



OpenArm + parallel gripper



Chest RGB



Wrist RGB

RGB + joint / gripper state

Flow policy + residual → action chunks

Recorded rollouts → offline refinement → autonomous evaluation

Scene images are teleoperated demonstration frames.

Place accurately across changing heights.

V01 / Autonomous placement

PLACEHOLDER · teleoperation still



STATIC PREVIEW · VIDEO RESERVED

Autonomous success



+40 percentage points

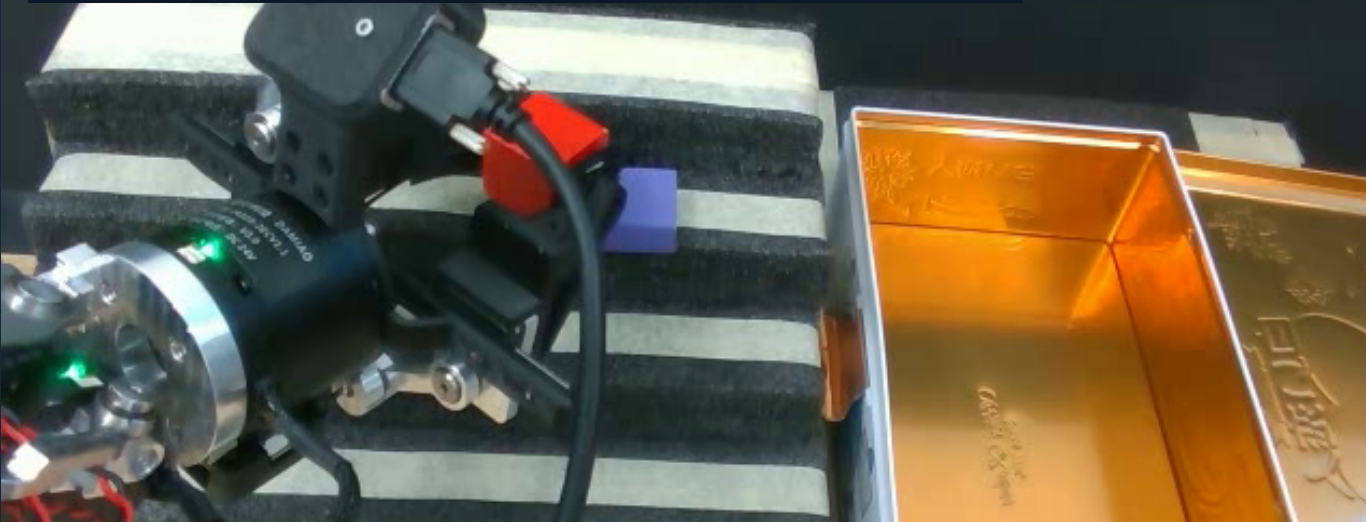
over the pretrained base

Place the toy fully inside the case.

Preserve the approach. Improve the finish.

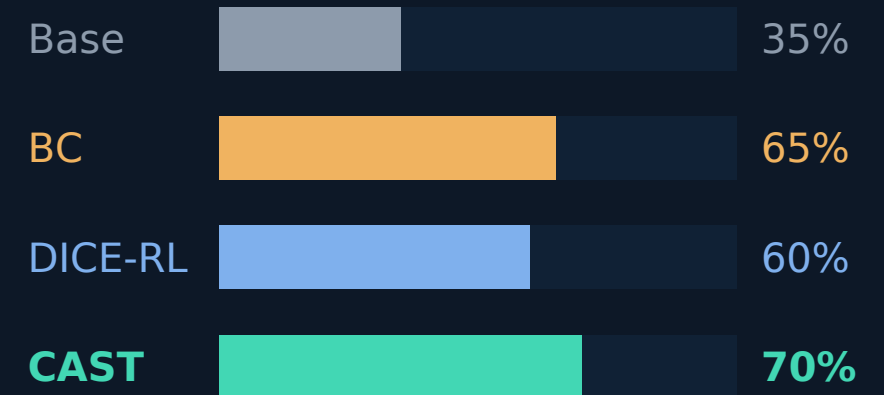
V02 / Autonomous stacking

PLACEHOLDER · teleoperation still



STATIC PREVIEW · VIDEO RESERVED

Autonomous success



+35 percentage points

over the pretrained base

A failed release does not make every earlier action wrong.

Improve the skill. Keep a persistent reference.

Critic guidance

Propose an improvement

Soft restoration

Retain a base reference

Bounded target steps

Moderate each correction

cast2027.github.io